

Chapter 3. **GROUPS**

3.3 Cyclic Groups

3.3.2 The Fundamental Theorem of Finite Cyclic Groups



Drawing the lattice diagram of the subgroups of a finite cyclic group

Example 3.35. Find all subgroups of the additive cyclic group $\mathbb{Z}_{100}$ and draw the lattice diagram.

$$\nearrow (\{0\})$$

Positive divisors of 100: 1, 2, 4, 5, 10, 20, 25
50 and 100 ($\mathbb{Z}_{100}$)

By the Fundamental Thm of Finite Cyclic Groups $\mathbb{Z}_{100}$ has 9 subgroups, one for each positive divisor of 100.

Each subgroup of $\mathbb{Z}_{100}$ is of the form $\underbrace{\langle \bar{d} \rangle}_{d \mathbb{Z}_{100}}$, where $d \mid 100$.

Notice that $\bar{d} = d \bar{1}$ and $\mathbb{Z}_{100} = \langle \bar{1} \rangle$.

Moreover, $|\langle \bar{d} \rangle| = o(d \bar{1}) = \frac{100}{d}$
Lemma 3.31

$$\underline{\text{Order } 2} : 2 = \frac{100}{d} \Rightarrow d = 50$$

$$\langle \overline{50} \rangle = 50 \mathbb{Z}_{100}$$

$$\underline{\text{Order } 4} : 4 = \frac{100}{d} \Rightarrow d = 25$$

$$\langle \overline{25} \rangle = 25 \mathbb{Z}_{100}$$

$$\underline{\text{Order } 5} : 5 = \frac{100}{d} \Rightarrow d = 20$$

$$\langle \overline{20} \rangle = 20 \mathbb{Z}_{100}$$

$$\underline{\text{Order } 10} : 10 = \frac{100}{d} \Rightarrow d = 10$$

$$\langle \overline{10} \rangle = 10 \mathbb{Z}_{100}$$

$$\underline{\text{Order } 20} : 20 = \frac{100}{d} \Rightarrow d = 5$$

$$\langle \overline{5} \rangle = 5 \mathbb{Z}_{100}$$

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Order 25: $25 = \frac{100}{d} \Rightarrow d = 4$

$$\langle \bar{4} \rangle = 4\mathbb{Z}_{100}$$

Order 50: $50 = \frac{100}{d} \Rightarrow d = 2$

$$\langle \bar{2} \rangle = 2\mathbb{Z}_{100}$$

$$\langle \bar{m} \rangle \subseteq \langle \bar{k} \rangle \Leftrightarrow k \mid m$$

$$\langle \bar{50} \rangle \subseteq \langle \bar{10} \rangle$$

$$\langle \bar{50} \rangle \subseteq \langle \bar{25} \rangle$$

$$\langle \bar{50} \rangle \subseteq \langle \bar{5} \rangle$$

$$\langle \bar{10} \rangle \subseteq \langle \bar{5} \rangle$$

$$\langle \bar{10} \rangle \subseteq \langle \bar{2} \rangle$$

